

scpviz: A Python bioinformatics toolkit for Single-cell Proteomics and multi-omics analysis

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Summary

Proteomics seeks to characterize protein dynamics by measuring both protein abundance and post-translational modifications (PTMs), such as phosphorylation, acetylation, and ubiquitination, which regulate protein activity, localization, and interactions. In bottom-up proteomics workflows, proteins are enzymatically digested into peptides that are measured as spectra, from which these peptide-spectrum matches (PSMs) are aggregated to infer protein-level identifications and quantitative abundance estimates. Analyzing the two levels of data at both the peptide level (short fragments observed directly) and the protein level (assembled from peptide evidence) in tandem is crucial for translating raw measurements into biologically interpretable results.

Single-cell proteomics extends these approaches to resolve protein expression at the level of individual cells or microdissected tissue regions. Such data are typically sparse, with many missing values, and are generated within complex experimental designs involving multiple classes of samples (e.g., cell type, treatment, condition). These properties distinguish single-cell proteomics from bulk experiments and create unique challenges in data processing, normalization, and interpretation. The single-cell transcriptomics community has established a mature ecosystem for managing similar challenges, exemplified by the scanpy package (Wolf et al., 2018) and the broader scverse ecosystem (Virshup et al., 2023). Building on these foundations, scpviz extends the AnnData data structure to the domain of proteomics, supporting a complete analysis pipeline from raw peptide-level data to protein-level summaries and downstream interpretation through differential expression, enrichment analysis, and network analysis. The core of scpviz is the pAnnData class, an AnnData-affiliated data structure specialized for proteomics. Together, these components make scpviz a comprehensive and extensible framework for single-cell proteomics. By combining flexible data structures, reproducible workflows, and seamless integration with the AnnData, scanpy and extended scverse ecosystem, the package enables researchers to efficiently connect peptide-level evidence to protein-level interpretation, thereby accelerating methodological development and biological discovery in proteomics.

Statement of need

Although general-purpose data analysis frameworks such as scanpy (Wolf et al., 2018) and the broader scverse ecosystem have become indispensable for single-cell transcriptomics, comparable tools for proteomics remain limited. Unlike transcriptomic data (counts, reads), proteomics measurements are inherently hierarchical. Peptide-level measurements provide the primary basis from which protein-level quantities are inferred, often with shared or ambiguous

peptide assignments and strong reliance on peptide confidence. In addition, missing data is pervasive in low-input and single-cell experiments, further complicating representation and analysis using flat feature-by-sample abstractions.

Existing proteomics software typically focuses on upstream tasks such as peptide identification and protein quantification, producing tabular outputs that are not designed for iterative downstream analysis. As a result, users must rely on ad hoc data structures to perform filtering, normalization, visualization, and interpretation, limiting reproducibility and making it difficult to integrate proteomics data into modern single-cell analysis workflows. These limitations are amplified in single-cell and spatial contexts, where complex experimental designs, sparse measurements, and the need to compare across multiple biological conditions require flexible data management and metadata-aware analysis.

scpviz addresses this gap by providing a unified framework for organizing and analyzing proteomics data from raw peptide-level evidence through protein-level summaries and biological interpretation. It is designed for computational biologists and proteomics researchers working with low-input or single-cell datasets generated by common analysis pipelines such as Proteome Discoverer or DIA-NN (Demichev et al., 2020). By enabling structured downstream analysis and integration with established single-cell ecosystems, scpviz supports reproducible and scalable proteomics workflows and facilitates cross-modality analyses that connect protein-level measurements to broader systems-level biology.

State of the field

A range of software tools exists for proteomics data processing and analysis, each addressing specific stages of the workflow. Upstream platforms such as Proteome Discoverer and DIA-NN (Demichev et al., 2020) provide peptide identification, protein inference, quantitative estimation, and built-in visualization capabilities. However, these environments are primarily designed around fixed analysis pipelines focused on spectral processing and quantification, and offer limited flexibility for downstream data manipulation, including user-defined normalization strategies, imputation methods, and iterative exploratory analysis. As a result, they typically export tabular outputs intended for external analysis rather than serving as extensible frameworks for single-cell or spatial proteomics workflows.

For single-cell proteomics specifically, several tools have been developed to address downstream analysis needs. The scp Bioconductor package (Vanderaa & Gatto, 2023) provides a comprehensive R-based framework for processing and analyzing mass spectrometry-based single-cell proteomics data, built on QFeatures and SingleCellExperiment. Pipeline-oriented tools such as the SCoPE2 pipeline, SPP (Scripts and Pipelines for Proteomics), QuantQC, and SCEPTRE offer lab-specific workflows for particular acquisition designs (e.g. isobaric carrier, plexDIA, nPOP). While these tools address important upstream and QC needs, they are generally pipeline-specific, R-based, or tied to particular experimental designs.

Meanwhile, the Python-based single-cell ecosystem continues to grow rapidly, with the scverse framework (Virshup et al., 2023) establishing AnnData as a community standard for interoperable single-cell analysis. Python-native proteomics algorithms such as directLFQ (Ammar et al., 2023) for label-free quantification and PIMMS (Webel et al., 2024) for missing value imputation are increasingly available. Most recently, ProteoPy (Fichtner et al., 2026) introduced an AnnData-based Python framework for bulk proteomics analysis – appearing as a preprint during the review period of this submission, and itself underscoring the timeliness of Python-native proteomics infrastructure. However, single-cell proteomics presents distinct challenges that bulk-oriented frameworks do not address: peptide-level information is critical not only for filtering (e.g. unique peptide thresholds, sparsity-aware filtering) but also for core quantification operations such as directLFQ normalization, which requires explicit peptide-protein relationships as input. Single-cell and spatial proteomics workflows therefore require a framework designed around these hierarchical relationships from the ground up, rather than

92 adapted from bulk assumptions. `scpviz` addresses this by maintaining paired peptide- and
93 protein-level `AnnData` objects linked by an explicit relationship matrix, enabling peptide-aware
94 operations throughout the analysis workflow within a unified Python framework compatible
95 with the broader `scverse` ecosystem.

96 While transcriptomics frameworks such as `scanpy` operate on a single level of abstraction
97 (gene-level counts), proteomics analyses depend critically on peptide-level measurements for
98 quantification, normalization, and confidence assessment, requiring extension to a hierarchical
99 structure that existing frameworks do not natively support. With this in mind, `scpviz` was
100 developed as a standalone framework to bridge this structural gap. Extending upstream
101 proteomics software would not address downstream, metadata-aware analysis needs and is
102 constrained in some cases by commercial software ecosystems, while embedding proteomics-
103 specific logic directly into transcriptomics frameworks would introduce unnecessary complexity
104 for transcriptomics use cases. By extending `AnnData` with a proteomics-specific data model
105 that explicitly captures peptide-protein relationships, `scpviz` enables proteomics data to be
106 analyzed within established single-cell workflows while preserving domain-specific rigor. This
107 positions `scpviz` as connective infrastructure between proteomics and single-cell analysis
108 ecosystems, rather than a replacement for existing tools.

109 Software design

110 The design of `scpviz` centers on the `pAnnData` class, an `AnnData`-affiliated data structure
111 specialized for proteomics. Rather than representing proteomics data as a single flat matrix,
112 `pAnnData` accounts for the hierarchical relationship between peptide-level and protein-level
113 measurements by pairing matched peptide (`.pep`) and protein (`.prot`) `AnnData` objects with
114 supporting attributes such as `.summary`, `.metadata`, `.stats`, and a protein-peptide relationship
115 (`.rs` matrix). This design allows users to preserve explicit peptide-protein relationships during
116 downstream analyses, enabling operations like peptide-level protein abundance normalization
117 or peptide-based fold-change aggregation for differential expression, while maintaining
118 compatibility with established Python libraries for data science and visualization (Figure 1).

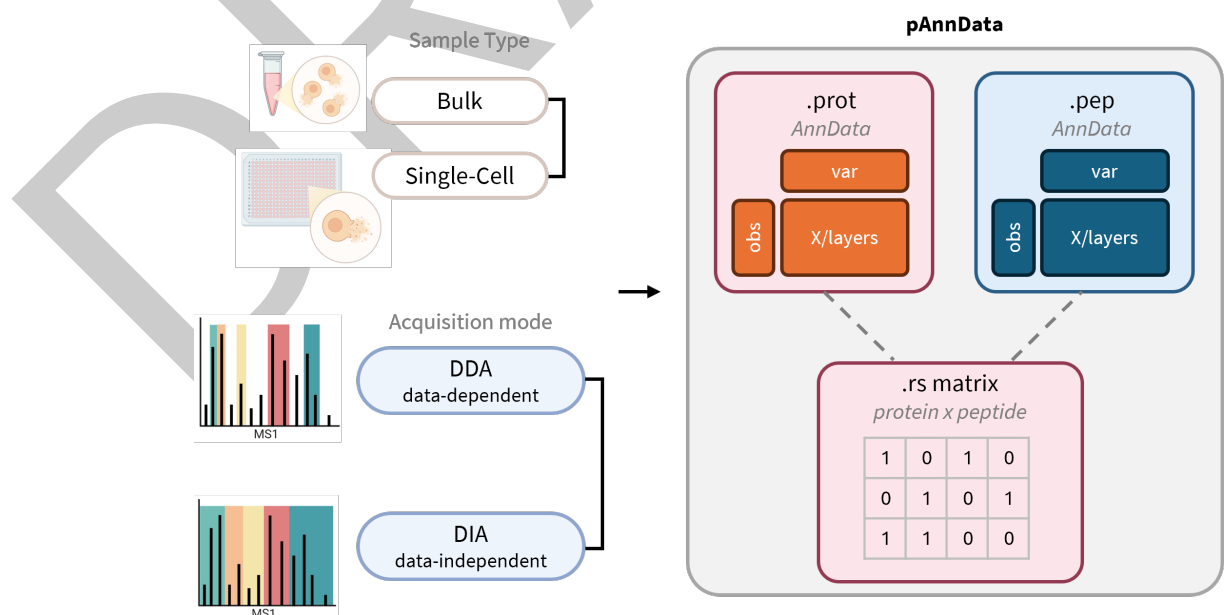


Figure 1: `pAnnData` integrates peptide and protein `AnnData` objects linked by a relationship matrix.

119 Proteomics-specific operations in `scpviz` are designed to operate uniformly across peptide

and protein-level data. By organizing functionality into mixin-based classes that act on underlying AnnData objects, common operations such as filtering, normalization, imputation, and summarization can be applied consistently to both peptides and proteins. This object-oriented design reduces code duplication while ensuring that shared analytical logic respects proteomics-specific constraints at each level of representation. Visualization and analysis utilities (e.g. PCA, UMAP, clustermaps, abundance plots) build directly on these structured representations (McInnes et al., 2018). For downstream interpretation, scpviz integrates external resources such as UniProt for annotation and STRING database for functional enrichment and network analysis (Snel et al., 2000; Szklarczyk et al., 2023), and incorporates proteomics-specific quantification strategies such as directLFQ (Ammar et al., 2023). By retaining AnnData compatibility, pAnnData objects can be used directly with tools such as scanpy (Wolf et al., 2018) and harmony (Korsunsky et al., 2019), enabling direct incorporation into established single-cell workflows.

Design decisions in scpviz emphasize separation between data representation and analytical operations. Core data structures encode peptide–protein relationships and metadata, while higher-level methods implement proteomics-aware transformations that can be composed, extended, or replaced. This modular architecture enables users to adapt the framework to evolving experimental designs without coupling downstream analysis logic to upstream data processing assumptions. The design philosophy of scpviz thus emphasizes both usability and extensibility. General users can rely on its streamlined API to import, process, and visualize single-cell proteomics data without deep programming expertise, while advanced users can extend the framework to accommodate custom analysis pipelines.

Research impact statement

scpviz has already been used in the analysis of multiple published studies and preprints across single-cell and bulk proteomics applications (Dutta, Pang, Coughlin, et al., 2025; Dutta, Pang, Donahue, et al., 2025; Pang et al., 2025; Uslan et al., 2025). In these works, the framework enabled structured downstream analysis of peptide and protein-level data, including differential expression, functional enrichment, and integration with transcriptomic measurements.

The development of scpviz has been shaped by close collaboration with staff scientists and users of the Proteome Exploration Laboratory (PEL) at Caltech, a shared mass spectrometry facility serving research groups across campus. Feedback from PEL staff and facility users directly informed the design and feature set of the package, ensuring that scpviz addresses practical needs across a range of single-cell and bulk proteomics workflows. In addition, scpviz has been incorporated into graduate-level training to demonstrate how proteomics workflows can be analyzed using pipelines common in single-cell transcriptomics analysis, lowering the barrier for new users entering the field.

The primary impact of scpviz lies in providing reusable infrastructure rather than task-specific analyses. By formalizing peptide–protein relationships within an AnnData-compatible data model, the package enables proteomics data to be analyzed using established single-cell tools for visualization, integration, and exploratory analysis. This structured representation supports reproducible downstream workflows and directly enables multi-omics integration. Because scpviz represents proteomics data as standard AnnData objects, proteomics and transcriptomics datasets can be jointly analyzed using the broader scverse ecosystem without custom data wrangling. This capability was used in practice in (Dutta, Pang, Coughlin, et al., 2025), where single-cell spatial proteomics data processed with scpviz was integrated with 10x scRNA-seq data from the Allen Brain Atlas mouse brain atlas. Both datasets were represented as AnnData objects, allowing direct cross-modality comparison of cortex-to-SNpc fold changes to benchmark competing normalization strategies, with directLFQ normalization showing substantially higher directional agreement between transcript ratios and protein log2 fold changes than median-based normalization with imputation. This cross-omics benchmarking approach using transcriptomic data as an orthogonal reference for evaluating

171 proteomics preprocessing was made tractable by the shared AnnData data model, and illustrates
172 the practical value of building proteomics infrastructure within an interoperable single-cell
173 ecosystem.

174 scpviz is released as open-source software with comprehensive documentation, automated
175 tests, and reproducible examples, supporting transparent and extensible research workflows.
176 Its emphasis on interoperability with widely used single-cell analysis libraries positions it
177 as infrastructure for continued method development as single-cell and spatial proteomics
178 technologies mature and scale.

179 AI usage disclosure

180 Generative AI tools were used during the development of this work to assist with code
181 refactoring, documentation drafting, and manuscript text editing. All software design decisions,
182 implementation, validation, and scientific interpretation were performed and reviewed by the
183 authors. No generative AI tools were used to generate or analyze research data, and all results
184 reported are reproducible from the publicly available source code and documentation.

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189 References

- 190 Ammar, C., Schessner, J. P., Willems, S., Michaelis, A. C., & Mann, M. (2023). Accurate
191 Label-Free Quantification by directLFQ to Compare Unlimited Numbers of Proteomes.
192 *Molecular & Cellular Proteomics*, 22(7). <https://doi.org/10.1016/j.mcpro.2023.100581>
- 193 Demichev, V., Messner, C. B., Vernardis, S. I., Lilley, K. S., & Ralser, M. (2020). DIA-
194 NN: Neural networks and interference correction enable deep proteome coverage in high
195 throughput. *Nature Methods*, 17(1), 41–44. <https://doi.org/10.1038/s41592-019-0638-x>
- 196 Dutta, S., Pang, M., Coughlin, G. M., Gudavalli, S., Roukes, M. L., Chou, T.-F., & Gradinaru, V. (2025, February 11). *Molecularly-guided spatial proteomics captures single-cell identity and heterogeneity of the nervous system*. <https://doi.org/10.1101/2025.02.10.637505>
- 199 Dutta, S., Pang, M., Donahue, R. R., Chou, T.-F., Seifert, A. W., & Gradinaru, V. (2025). *Parkinson's disease modeling in regenerative spiny mice (Acomys dimidiatus) captures key disease-relevant behavioral, histological, and molecular signatures* (p. 2025.11.06.687049).
202 bioRxiv. <https://doi.org/10.1101/2025.11.06.687049>
- 203 Fichtner, I. D., Temesvari-Nagy, L., Sahm, F., Gerstung, M., & Bludau, I. (2026, April
204 1). *ProteoPy: An AnnData-based framework for integrated proteomics analysis*. <https://doi.org/10.64898/2026.03.31.715273>
- 206 Korsunsky, I., Millard, N., Fan, J., Slowikowski, K., Zhang, F., Wei, K., Baglaenko, Y., Brenner, M., Loh, P., & Raychaudhuri, S. (2019). Fast, sensitive and accurate integration of
207 single-cell data with Harmony. *Nature Methods*, 16(12), 1289–1296. <https://doi.org/10.1038/s41592-019-0619-0>
- 210 McInnes, L., Healy, J., Saul, N., & Großberger, L. (2018). UMAP: Uniform Manifold
211 Approximation and Projection. *Journal of Open Source Software*, 3(29), 861. <https://doi.org/10.21105/joss.00861>

- 213 Pang, M., Jones, J. J., Wang, T.-Y., Quan, B., Kubat, N. J., Qiu, Y., Roukes, M. L., & Chou,
214 T.-F. (2025). Increasing Proteome Coverage Through a Reduction in Analyte Complexity
215 in Single-Cell Equivalent Samples. *Journal of Proteome Research*, 24(4), 1528–1538.
216 <https://doi.org/10.1021/acs.jproteome.4c00062>
- 217 Snel, B., Lehmann, G., Bork, P., & Huynen, M. A. (2000). STRING: A web-server to retrieve
218 and display the repeatedly occurring neighbourhood of a gene. *Nucleic Acids Research*,
219 28(18), 3442–3444. <https://doi.org/10.1093/nar/28.18.3442>
- 220 Szklarczyk, D., Kirsch, R., Koutrouli, M., Nastou, K., Mehryary, F., Hachilif, R., Gable, A.
221 L., Fang, T., Doncheva, N. T., Pyysalo, S., Bork, P., Jensen, L. J., & von Mering, C.
222 (2023). The STRING database in 2023: Protein-protein association networks and functional
223 enrichment analyses for any sequenced genome of interest. *Nucleic Acids Research*, 51(D1),
224 D638–D646. <https://doi.org/10.1093/nar/gkac1000>
- 225 Uslan, T., Quan, B., Wang, T.-Y., Pang, M., Qiu, Y., & Chou, T.-F. (2025). In-Depth
226 Comparison of Reagent-Based Digestion Methods and Two Commercially Available Kits
227 for Bottom-Up Proteomics. *ACS Omega*, 10(10), 10642–10652. <https://doi.org/10.1021/acsomega.4c11585>
- 229 Vanderaa, C., & Gatto, L. (2023). The Current State of Single-Cell Proteomics Data Analysis.
230 *Current Protocols*, 3(1), e658. <https://doi.org/10.1002/cpz1.658>
- 231 Virshup, I., Bredikhin, D., Heumos, L., Palla, G., Sturm, G., Gayoso, A., Kats, I., Koutrouli,
232 M., Berger, B., Pe'er, D., Regev, A., Teichmann, S. A., Finotello, F., Wolf, F. A., Yosef,
233 N., Stegle, O., & Theis, F. J. (2023). The scverse project provides a computational
234 ecosystem for single-cell omics data analysis. *Nature Biotechnology*, 41(5), 604–606.
235 <https://doi.org/10.1038/s41587-023-01733-8>
- 236 Webel, H., Niu, L., Nielsen, A. B., Locard-Paulet, M., Mann, M., Jensen, L. J., & Rasmussen,
237 S. (2024). Imputation of label-free quantitative mass spectrometry-based proteomics
238 data using self-supervised deep learning. *Nature Communications*, 15(1), 5405. <https://doi.org/10.1038/s41467-024-48711-5>
- 240 Wolf, F. A., Angerer, P., & Theis, F. J. (2018). SCANPY: Large-scale single-cell gene expression
241 data analysis. *Genome Biology*, 19(1), 15. <https://doi.org/10.1186/s13059-017-1382-0>